Fiscal Year:	FY 2009	**Task Last Updated:**	FY 09/04/2009
PI Name:	Kanas, Nick M.D.		
Project Title:	Effects of High vs. Low Autonomy on Space Crewmember Performance (Autonomy)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	415-750-2072
Organization Name:	University of California-San Francisco/Northern California Institute for Research & Education		
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City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/12/2005	**End Date:**	03/31/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Shea, Cami1e	**Contact Phone:**	281-244-2017
Contact Email:	shea@dsls.usra.edu		
Flight Program:			
Flight Assignment:	NOTE: End date should be 3/31/2010, per C. Guidry/JSC (4/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Boyd, Jennifer (Department of Psychiatry, UCSF) Marmar, Charles (Department of Psychiatry, UCSF) Weiss, Daniel (Department of Psychiatry, UCSF) Neylan, Thomas (Department of Psychiatry, UCSF)		
Grant/Contract No.:	NNJ05HG01A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	During future space missions to the Moon or Mars, crewmembers will have more autonomy in planning and conducting their activities as a result of the long distances involved and the tasks that will be needed to explore a planetary surface. The impact of increased autonomy on crew performance has not been evaluated formally. In past missions, strict adherence to timelines has been reported to be fatiguing and stressful for astronauts, and this can affect interpersonal interactions, resulting in increased team conflict and reduced morale and productivity. Our long term goal is to study low versus high autonomy in space on-board the ISS. Our short term goal is to conduct a series of flight feasibility pilot studies involving space simulation environments on Earth to demonstrate the safety of high autonomy on isolated crews and to test the usefulness of measures of crewmember mood, interpersonal interactions, and work performance in low and high autonomy conditions. The results of these studies will help up develop hypotheses for a future ISS study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of our pilot studies will help us in planning future studies of low versus high autonomy in space on-board the ISS. The results also will translate to work groups on Earth and the effectiveness of increased group autonomy on work place mood, interpersonal interactions, and performance. Many of our variables, such as displacement of dysphoria to outside groups, task versus emotional leadership characteristics, and the effect of national and organizational culture on worker well-being and performance, have obvious implications for national and international working groups on Earth.
Task Progress:	To date, we have studied three space simulation environments: NEEMO 12 and 13, HMP 2008, and (under a different contract) the Russian 500 105-day simulation. In October, 2009, we will study another NEEMO mission (NEEMO 14). We are gradually accumulating information suggesting that isolated crewmembers like and tolerate high autonomy conditions. Despite some cultural differences, crews perform well and accomplish mission goals. Mission control subjects report some confusion concerning their work role during high crew autonomy conditions. Our results are largely descriptive due to the low numbers of subjects, but as we accumulate more information, we may be able to employ standard statistical techniques to our data set. In addition, we have piloted some new measures of autonomy and work performance, which we plan to use in future on-orbit studies.
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